

Business Problem-Solving Case Studies

Strategic Procurement | Supplier Development | Cost Optimization | Material Planning | Supply Continuity

Professional Profile: Strategic Procurement & Supply Chain professional with 11+ years of experience across Electrical Equipment, Motors & Alternators, Rolling Stock/Railways and Automotive manufacturing. This portfolio presents anonymized examples of the problem-solving approach used across procurement, planning, supplier management and manufacturing supply chains.

How I Solve Supply-Chain Problems

- 1. Detect: identify the business risk using demand, MRP, inventory, supplier and delivery data.
- 2. Diagnose: separate the root cause from the immediate symptom using structured analysis.
- 3. Strategize: define sourcing, supplier, inventory or planning actions based on cost, quality, delivery and risk.
- 4. Execute: align suppliers and cross-functional stakeholders and establish clear ownership and timelines.
- 5. Control: track KPIs, corrective actions and standard work to prevent recurrence.

CASE STUDY 1

Supplier Delivery & OTD Improvement

Context: Procurement and supplier management for critical components used in electric motors and alternators.

Business Problem: Supplier delivery commitments were not consistently aligned with changing MRP demand and production priorities, creating potential shortage and line-continuity risk.

My Approach

- Segmented requirements into back orders, live orders and future demand to identify critical supply exposure.
- Compared MRP requirements with supplier commitments, lead times and available capacity.
- Established structured supplier reviews covering OTD/OTIF, open orders, capacity and recovery actions.
- Prioritized critical components and escalated constraints with clear ownership and due dates.
- Used supplier scorecards and recurring reviews to move from reactive follow-up to performance management.

Business Outcome

- Improved visibility of supplier commitments and critical shortages.
- Created a structured mechanism for early identification and escalation of supply risk.
- Strengthened supplier accountability through KPI-based performance reviews.

KPI Evidence :

KPI	
Supplier OTD/OTIF	actual improvement (%)
Critical shortages	monthly shortage trend before/after
Past-due orders	Reduction in past-due quantity or value
Recovery lead time	Average days required to recover critical gaps

Skills Demonstrated: Supplier Performance Management | Supply Risk Management | MRP | Supplier Development | Stakeholder Management

Procurement Cost Optimization & Supplier Negotiation

Context: Strategic procurement of electrical/mechanical components and support for prototype/new-component costing.

Business Problem: Cost competitiveness required stronger visibility of material, process and supplier cost drivers before negotiation and sourcing decisions.

My Approach

- Collected RFQ and supplier cost information and separated material, conversion, tooling, logistics and other relevant cost elements.
- Used should-cost thinking and cost-driver analysis to challenge supplier quotations.
- Benchmarked supplier offers and identified negotiation levers without compromising technical requirements.
- Engaged Engineering and suppliers to evaluate feasible cost-reduction and value-engineering opportunities.
- Documented commercial assumptions and action items for sourcing decisions.

Business Outcome

- Improved transparency of quotation and cost drivers.
- Created a fact-based negotiation approach rather than relying only on historical prices.
- Identified opportunities for cost reduction and improved commercial governance.

KPI Evidence:

KPI	
Annualized savings	Validated savings value / %
Cost reduction	Negotiated reduction versus baseline
RFQ coverage	Number/value of components sourced or re-quoted
Cost-driver visibility	Categories where should-cost analysis was applied

Skills Demonstrated: Strategic Sourcing | Should Costing | Cost Optimization | RFQ Management | Negotiation

Material Shortage & Supply-Risk Resolution

Context: High-mix manufacturing environment with multiple component categories, supplier lead times and production priorities.

Business Problem: A material shortage or supplier constraint could disrupt production, particularly where lead times were long or supply was dependent on a limited supplier base.

My Approach

- Reviewed demand, MRP, stock, open orders, supplier commitments and lead-time position to quantify the exposure.
- Classified the issue by urgency, production impact and recovery time.
- Developed a recovery plan covering supplier expediting, prioritization, alternate actions and internal alignment.
- Coordinated Production, Planning, Procurement, Quality and suppliers around a single recovery plan.
- Captured root cause and preventive actions to reduce repeat shortages.

Business Outcome

- Reduced reactive firefighting by using structured shortage prioritization.
- Improved cross-functional visibility of supply risk and recovery ownership.
- Strengthened preventive supply planning and escalation discipline.

KPI Evidence:

KPI	
Line-stop risk	Number of avoided/mitigated incidents
Shortage recovery	Average recovery time before/after
Critical parts	Number of parts covered by risk review
Repeat shortages	Trend after corrective action

Skills Demonstrated: Supply Continuity | Root Cause Analysis | Risk Management | Material Planning | Cross-functional Leadership

Inventory Optimization & MRP Discipline

Context: Materials planning and inventory management across high-volume automotive and electrical-equipment manufacturing environments.

Business Problem: The business needed to balance production continuity against excess inventory, long lead times, demand variability and supplier constraints.

My Approach

- Reviewed demand, lead time, stock position, open orders and consumption patterns to identify inventory drivers.
- Used MRP outputs to prioritize replenishment and distinguish genuine requirements from timing or parameter issues.
- Aligned safety-stock and replenishment decisions with lead time, criticality and supply risk.
- Worked with suppliers and internal stakeholders to improve commitment accuracy and reduce unnecessary inventory exposure.
- Tracked inventory and shortage indicators to maintain the balance between availability and working capital.

Business Outcome

- Improved visibility of inventory drivers and replenishment priorities.
- Supported production continuity while identifying opportunities to reduce excess stock.
- Strengthened MRP-based planning discipline and decision-making.

KPI Evidence:

KPI	
Inventory value	Validated reduction / avoided increase
Days of inventory	Before/after DOH
Inventory turns	Before/after turns
Material availability	Service level / shortage metric

Skills Demonstrated: Inventory Optimization | MRP | Material Planning | Demand Planning | Working Capital

NPI / BOM / ECN & Supplier Coordination

Context: Procurement and supply-chain support for prototype development, new components and engineering changes in electrical equipment.

Business Problem: Engineering changes and new products can create commercial, technical and supply risks when supplier understanding, BOM requirements and implementation timing are not aligned.

My Approach

- Reviewed BOM/ECN requirements with Engineering and suppliers and identified material, technical and commercial implications.
- Coordinated technical clarification calls to close supplier questions before implementation.
- Supported prototype development and cost analysis for new components.
- Aligned supplier readiness, lead time and procurement actions with implementation timelines.
- Tracked open actions across Engineering, Procurement and suppliers to reduce execution gaps.

Business Outcome

- Improved cross-functional visibility during NPI and engineering-change implementation.
- Reduced ambiguity between technical requirements and supplier execution.
- Strengthened supplier readiness and procurement involvement in product changes.

KPI Evidence:

KPI	
NPI cycle time	Before/after timeline if measured
ECN closure	Average closure time / open-item reduction
Supplier readiness	% or number of suppliers/components ready on time
Prototype cost	Cost variance versus target if validated

Skills Demonstrated: NPI | BOM Management | Engineering Change Management | Supplier Collaboration | Cost Analysis

Portfolio Summary

Business problems I solve: supplier delivery risk, cost pressure, material shortages, inventory imbalance, supplier capability gaps and NPI/engineering-change execution.

Leadership approach: data → root cause → strategy → stakeholder alignment → supplier action → KPI control.

Core capability: connecting Procurement, Planning, Engineering, Production, Quality, Finance and suppliers to deliver balanced outcomes across Cost, Quality, Delivery, Inventory and Risk.